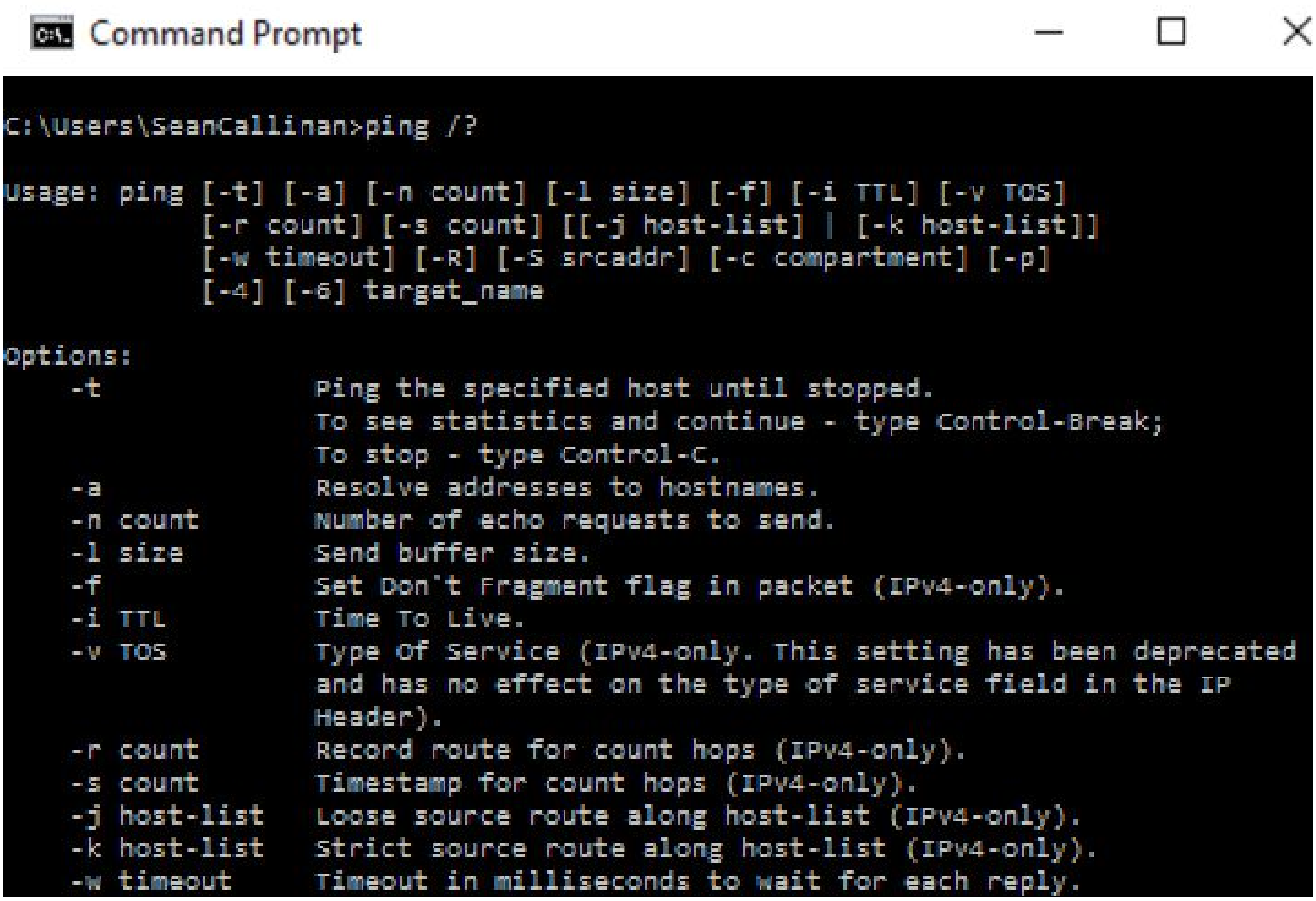
**************WINDOWS COMMAND PROMPT**************

Command Prompt —Command Prompt, or CMD.exe (or just CMD) is a Command-Line Interpreter for Windows Operating Systems. This interface allows a user OR an admin to directly communicate to the operating system using a TEXT BASED interface. If you are making system level changes, you will need to run as Administrator. This windows lets you speak DIRECTLY to the operating system OR network! Microsoft command-line tools are command lines specific to Microsoft operating systems. You will need to know the appropriate command-line tool when given a scenario.

What is SYNTAX? **Syntax of a computer language is the set of rules that defines the combinations of symbols that are considered to be a correctly structured document or fragment in that language. So when I say "syntax", I'm REALLY saying "The actual EXACT words or characters you'll enter to get a certain response from CMD.**

What is a SWITCH in CMD? A SWITCH is an optional command that can be given after the main commands listed below. For example…. This screenshot shows ping /?...... You can see that the switches such as "-t" and "-a", if used after ping, will alter the basic ping command to do something more specific.



```
Command Prompt

C:\Users\SeanCallinan>ping /?

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
```

/? : This command is also known as the HELP command. When placed after any command, it will give a help list of how to use that specific command! This is placed AFTER a command. Example: cd /?

Help: The same as /?, but it will be placed BEFORE a command. Example: help cd

Navigation

Navigation using the command line allows the user to navigate the file system using predetermined commands.

WHAT IS A DIRECTORY? **Don't be fooled! Directory is just a fancy word for a FOLDER…. And eventually we'll talk about how it refers to a folder and all of its subfolders, or folders within folders! FOLDER=DIRECTORY!**

cd —The **cd** command allows the user to change directories and is shorthand for the **chdir** command.
dir — The **dir** command allows the user to view a listing of files and folders/subdirectories in a directory.
md —The **md** command allows the user to make a directory and is shorthand for the **mkdir** command.
rmdir —The **rmdir** command allows the user to delete directories. The **rd** command is shorthand for this.
del —No shenanigans with this one. It means delete.

Drive Navigation Inputs

The **cd** command allows for navigation within the current drive. To navigate to another drive, you may use the drive letter followed by a semicolon command.

- **C:** — Changes to C drive
- **D:** — Changes to D drive
- **X:** — Changes to X drive

Command-Line Tools

Command-line tools can also be used for numerous other functions, such as testing or tracing network connectivity or paths. You should be familiar with common command-line tools.

hostname —The **hostname** command is used to pull up the identity of the computer the Command Prompt is open on.
winver —The **winver** command is used to display a GUI dialog box displaying the Windows version and edition
shutdown —The **shutdown** command can be used for scheduling a complete shutdown or a restart remotely or locally.
exit — Closes command prompt.

ipconfig — The **ipconfig** command is used to display basic connectivity information, such as the IP address, subnet mask, and default gateway. This command is highly useful in diagnosing network issues. It will show most of the network information you'll need.

NOTE**** There are a few specific IPCONFIG commands you'll need to know…………

ipconfig /all - not only will ipconfig /all show you your basic network settings, it will also show you mac address, detailed ipv6 information, ip address leases and dhcp connection information!
ipconfig /release (or release6) - this command will release your ip address back into the scope.
ipconfig /renew (or renew6) - this command will select your new ip address from the scope.
ipconfig /flushdns - **this command will clear the dns resolver cache on that specific machine! remember that when you first connect to a fqdn, a dns server is contacted to resolve that web address to an ip address that the computer can understand. this data is saved in the dns resolver cache to make future connections to that ip address faster! a very common series of commands to resolve connection is ipconfig /release, then ipconfig /renew, then ipconfig /flushdns . try it**

ping —Ping will send a specific amount of packets to a destination in order to test or VERIFY CONNECTIVITY
pathping —The **pathping** command is a mixture of the **tracert** and **ping** commands.
tracert —The **tracert** command is used to show the path a packet takes on a network to arrive at a

specified destination.
netstat — The **netstat** command is used to display all the listening and established connections on the host network.
nslookup —The **nslookup** command is used to verify DNS addresses. This command can be used for inline query or interactively.

chkdsk —The **chkdsk** command is used to view information about the hard disk, including the creation and viewing of reports, as well as to correct file system problems and disk errors. Note* CHKDSK /F will scn and FIX any errors! You'll use this as a tech!
diskpart —The **diskpart** command is a tool for managing disks, partitions, and volumes.
format — The **format** command is used to remove data from disks and prepare disks for new use. It will format a partition to a selected file system. File system examples: FAT16, FAT32, FAT64 (exFAT), NTFS, CDFS.

SO…… these next three are copy commands… BUT…. They have a few small differences we must know!~
copy —This command is a simple copy command, used for files
xcopy —Will copy a specified files, AND if desired, a directory tree to a specified destination.
robocopy —ROBOCOPY stands for ROBUST COPY FOR WINDOWS. Robocopy will copy files and directories, as well as file attributes, timestamps and other small details. ROBOCOPY also offers more switches than COPY or XCOPY.

net user — The **net user** command is a subcommand used to list all local accounts on the host system.
net use — The **net use** command is used to map drive letters to network shares.
gpuupdate — This command updates local and Active Directory group policy settings. (VERY similar to the GPEDIT command.)
gpresult — This command DISPLAYS group policy settings.
bootrec — This command will be used to troubleshoot the operating system IF startup repair doesn't work!. One VERY COMMON command is BOOTREC /FIXBOOT, which is the typical command to fix an error such as "BOOTMGR is missing."
sfc — SYSTEM FILE CHECKER! This command will be your best friend when attacking errors in WINDOWS SYSTEM FILES specifically. **A common command used by techs is SFC /SCANNOW, which will basically scan an entire system for Windows file errors and repairs them.**

SOME EXTRAS:

Microsoft Management Console (MMC) Snap-In
The Microsoft Management Console (MMC) snap-in is a GUI for running administrative and configuration tools.
Event Viewer (**eventvwr.msc**)—Event viewer is used to view application error logs, system errors, and security audit records.
Disk Management (**diskmgmt.msc**)—Disk Management is used to view disk information on the physical disk and the formatted file systems it contains.
Task Scheduler (**taskschd.msc**)—Task Scheduler allows you to configure automated tasks that will run on a schedule at specified times.
Device Manager (**devmgmt.msc**)—Device Manager allows you to view the status of devices, view the properties, and modify the configuration parameters.
Certificate Manager (**certmgr.msc**)—Certificate Manager is used to manage and view the certificates used by the OS and web browser.
Local Users and Groups (**lusrmgr.msc**)—As a system administrator, you need to know how to create and delete users and maintain their accounts, as well as how to establish secure passwords.

Performance Monitor (**perfmon.msc**)—Performance Monitor displays in real time how the computer uses memory, disk, CPU, and the network, to help diagnose performance issues.
Group Policy Editor (**gpedit.msc**)—Group Policy Editor is used to edit the local group policy for the OS.
System Information (**msinfo32.exe**)—System Information displays advanced hardware and driver information. Useful for viewing data about the computer and can be exported to a text document.
Resource Monitor (**resmon.exe**)—Resource Monitor shows how resources are being utilized by the CPU, memory, disk, and network.
System Configuration (**msconfig.exe**)—System Configuration can be run by the **msconfig** command. It allows you to change how Windows boots and what programs start with Windows.
Disk Cleanup (**cleanmgr.exe**)—Disk Cleanup frees up space while not affecting the integrity of the files.
Disk Defragment (**dfrgui.exe**)—Disk Defragment relocates pieces of large files to continuous space on a disk for optimized performance.
Registry Editor (**regedit.exe**)—Registry Editor allows the user to view and change the Windows registry. The registry contains some operating system and application settings that may need to be changed in advanced troubleshooting.

tasklist — This command creates a list of applications and services currently running on a system. This command will ALSO give the PID, or process ID number of a process or service!

taskkill — This command is usually used to end a process using the PID!